

The empirical recurrence for OEIS A250960: recovering a conjecture that is not written down

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Abstract

OEIS A250960 records “Empirical recurrence of order 75”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 75, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 75 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250960 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with no 2×2 subblock having its minimum diagonal element less than its minimum antidiagonal element.”. It has offset 1 and begins

14423, 768855, 42534386, 2395598690, 134817198768, 7555575891628, 422498268460458, 23613315121132337, ...

The entry states:

Empirical recurrence of order 75 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:54:26, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 75 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{75} c_i a(n-i),$$

c_1	256	c_{20}	−271965013620126868513582577	c_{39}	4557761931965286650
c_2	−30840	c_{21}	1573847254049184279612954823	c_{40}	−728158534751425947
c_3	2357917	c_{22}	−8414945407260527661454525223	c_{41}	1093771219540156515
c_4	−129427572	c_{23}	41655404363222909353290600679	c_{42}	−1544775957860077587
c_5	5456092477	c_{24}	−191263074778713501722716750491	c_{43}	2051262121048722090
c_6	−184317373169	c_{25}	815951391136651285708491163304	c_{44}	−256056957807532436
c_7	5138602475396	c_{26}	−3239195028960037486764612187562	c_{45}	3004127994408011091
c_8	−120810328888253	c_{27}	11982697420574659758705659006376	c_{46}	−3311579570370237408
c_9	2435315828773387	c_{28}	−41358584980628984780359357229124	c_{47}	3428585800937148520
c_{10}	−42649867567085336	c_{29}	133342140107480818900858532125998	c_{48}	−333229277600790381
c_{11}	655892304772787582	c_{30}	−401984023970540741142791562685726	c_{49}	3038496940621543526
c_{12}	−8935793997873792029	c_{31}	1134210698488530141481600517481808	c_{50}	−259747570456575666
c_{13}	108650076342181524613	c_{32}	−2997670477074219972352411063623468	c_{51}	2079968019238100124
c_{14}	−1186434987116514641902	c_{33}	7426784836595383758235277198115792	c_{52}	−155865660574868748
c_{15}	11697816901147142825470	c_{34}	−17259495009059148359143680161263028	c_{53}	1091801116370184960
c_{16}	−104623008473553568652654	c_{35}	37645460763572398365415884675691441	c_{54}	−713959770092983899
c_{17}	852253058198578161446800	c_{36}	−77102108412970344457789683337381889	c_{55}	4352082748706867254
c_{18}	−6345630398978186322123074	c_{37}	148342860084206318992723364157321092	c_{56}	−246875149741643603
c_{19}	43322715399713041389695473	c_{38}	−268199537017046565436370344124529508	c_{57}	1300686417108529586

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 75, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $75 + 4$ terms removed returns order 75 as well, so $\deg q_{\min} = 75 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $75 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 75 that this sequence satisfies — and that family turns out to have exactly one member.

References

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